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class test

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1) Image enhancement in spatial domain

Image enhancement in the spatial domain improves the quality of an image by directly modifying the pixel intensity values. The main objective is to improve contrast, remove noise.

(a) Gray-level transformation

these modify pixel intensities using mathematical eqn.

Types

- * Image negative
- * Log transformation
- * Power-law
- * Contrast stretching

Advantages

- * Simple and computationally efficient
- * Improves brightness and contrast

Applications

- * Medical imaging
- * Satellite images

(b) Histogram processing

A histogram represents the distribution of pixel intensities

Techniques

- * Histogram equalization
- * Histogram specification
- * Histogram stretching

Advantages

- * Enhances image contrast
- * Reveals hidden details

Applications

- * medical imaging
- * Remote sensing

(C) Spatial Filtering

uses a filter mask (kernel) that moves across the image

Types

- * smoothing filter
- * sharpening filter

Advantages

- * Reduce noise
- * Highlights important features

Applications

- * Face recognition
- * OCR

Justification

- * Gray-Level Transformations
- * Histogram processing
- * spatial Filtering

2)

Image smoothing	Image sharpening
* removes noise * Blurs image slightly * uses low-pass filters * Low-pass filters * Reduces high-Frequency components * suitable for noisy images	* enhances edges * makes details more clear * uses high-pass filters * High pass filters * enhance high-Frequency components * Suitable for edge enhancement

Spatial Smoothing Filters

Working principle

Replace each pixel with the average & average of neighbour pixels

Types

- * Mean Filter
- * Gaussian Filter
- * Median Filter

Advantages

- * Removes Gaussian & salt and pepper noise
- * Produces smoother image
- * Applications
 - * Medical Images
 - * CCTV
 - * Photography

Spatial Sharpening Filter

Working Filter

enhance rapid intensity changes using image

Types

- * Laplacian Filter
- * Sobel Filter
- * Prewitt Filter

Advantages

- * Object detection
- * Improves details

Applications

- * Fingerprint recognition
- * Industrial inspection

3)

Frequency Domain Enhancement

Instead of modifying pixels directly, the image is transformed into the frequency domain using the Fourier Transform (FFT)

Process

- * Convert image to frequency domain
- * Uses the FFT
- * Apply frequency filter
- * Perform inverse FFT
- * Obtain enhanced image

Two types

- * LPF
- * HPF

steps
 image $\rightarrow$ Fourier transform $\rightarrow$ Frequency Filters $\rightarrow$
 inverse Fourier transform $\rightarrow$ enhanced image
 comparison

Low-pass Filter	High-pass Filter
removes noise	enhances edges
produces blur	produces sharp images
remove high frequency	removes Low Freq
used for smoothing	used for sharpening

4) image segmentation
 image segmentation divides an image into a meaningful region or objects for easier analysis

Techniques

- 1) point detection
- 2) detect isolated pixels

Advantages

- simple
- Fast

Limitations

- defect detection

Techniques

point detection

edge detection

thresholding

region-based

Advantages

- simple
- accurate edges
- complete region
- more suitable technique
- edge detection accurately identifies object boundaries
- region based segmentation groups related pixels into complete objects
- together they provide better segmentation accuracy

a) Edge detection

Edge detection identifies sudden changes in image intensity

Applications

- Face recognition
- Traffic sign detection
- Medical imaging

b) Edge Linking

Edge Linking connects broken edges segments into continuous boundaries

methods

- Hough transform
- Graph-based methods

Applications

- * shape recognition
- * object tracking

c) Boundary detection

Boundary detection identifies the complete outer boundary of an object

Importance

- * measures object size
- * Find the object shape
- * Helps in classification

Applications

- * medical diagnosis
- * OCR

Contribution to Object Recognition

- * edge detection
- * edge linking
- * Boundary detection

Simple diagram

original image



Image enhancement



edge linking



Boundary detection



Object recognition